

Tema 4

Redes neuronales convolucionales

REDES NEURONALES Y DEEP LEARNING



Universidad
Internacional
de Valencia

Prof: Luis Márquez Carpintero

En el día de hoy

- Recordatorio
- Tipos de redes
- Clip

Encuesta previa

- ¿Cómo pensáis que Google Fotos busca en imágenes?

Recordemos

Imagen

1	1	1	0	0
0	1	1	1	0
0	0	1	1	1
0	0	1	1	0
0	1	1	0	0

Kernel

1	0	1
0	1	0
1	0	1

1 _{x1}	1 _{x0}	1 _{x1}	0	0
0 _{x0}	1 _{x1}	1 _{x0}	1	0
0 _{x1}	0 _{x0}	1 _{x1}	1	1
0	0	1	1	0
0	1	1	0	0

Image

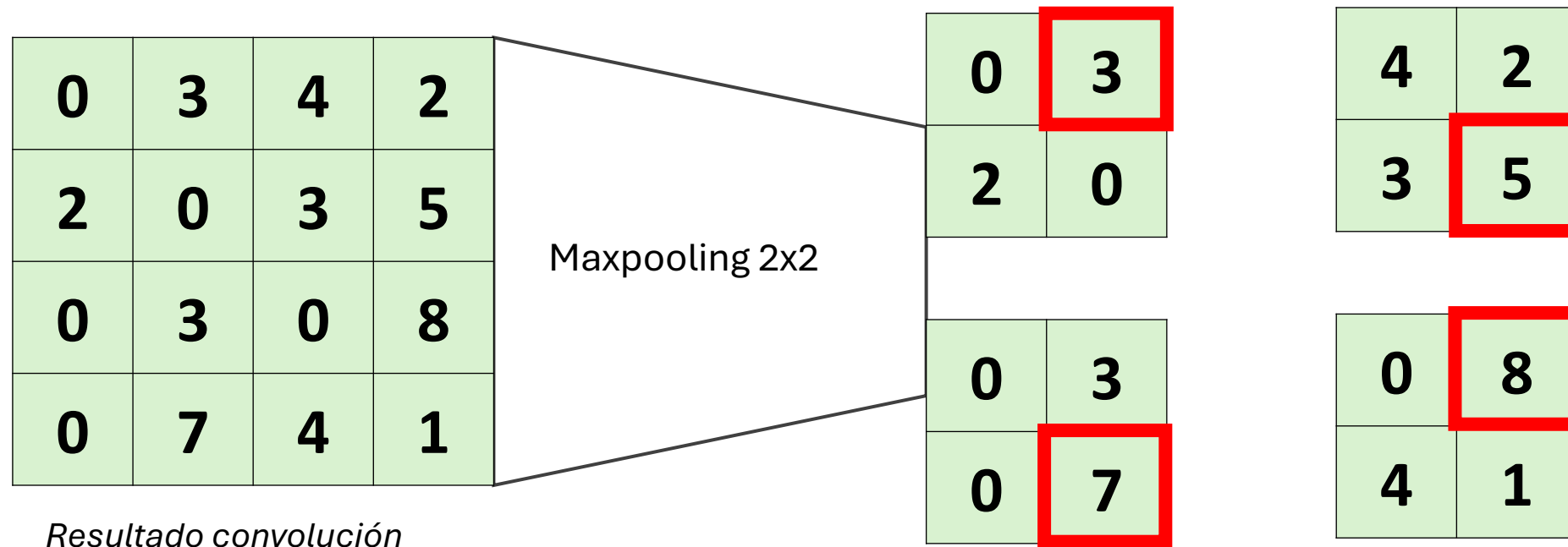
Output

4		

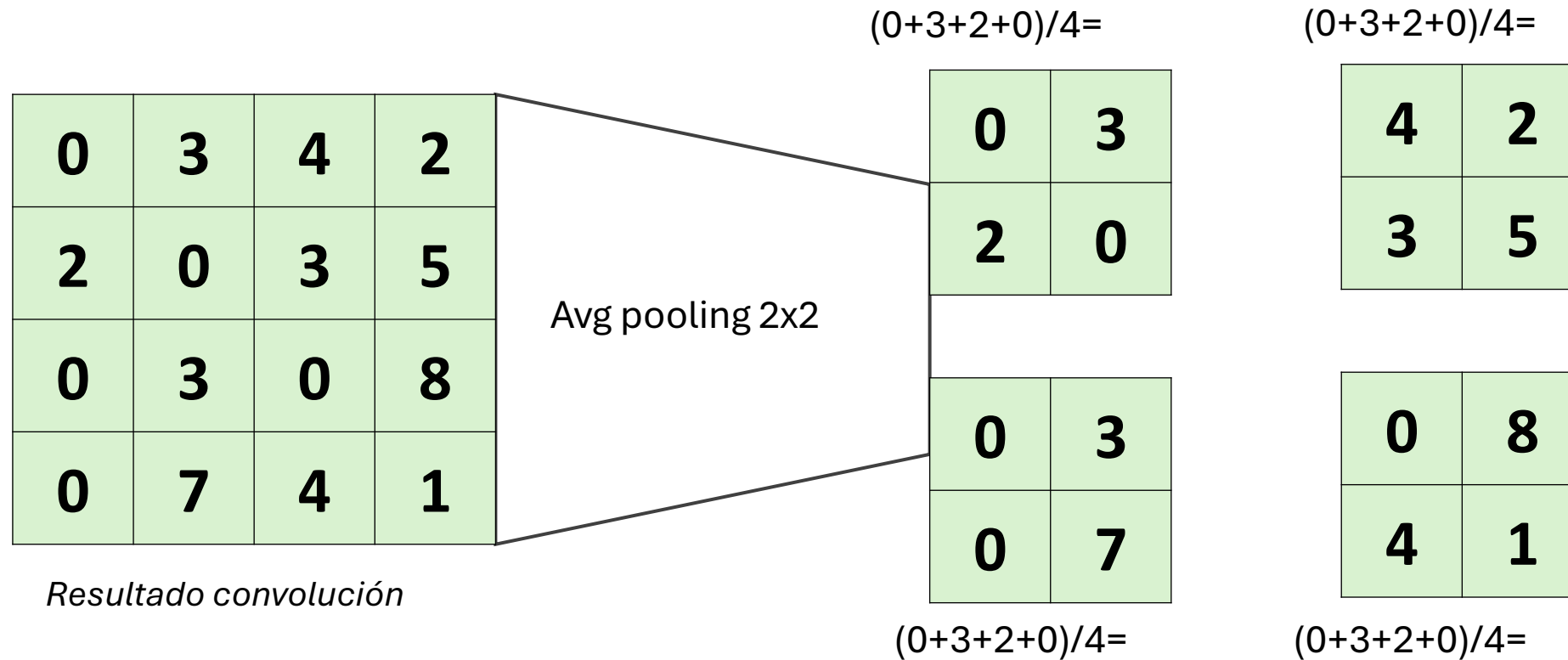
Convolved
Feature

<https://glassboxmedicine.com/2019/07/26/convolution-vs-cross-correlation/>

Recordemos



Recordemos

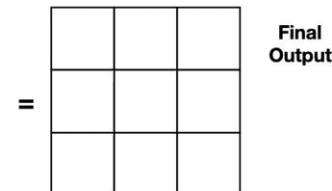
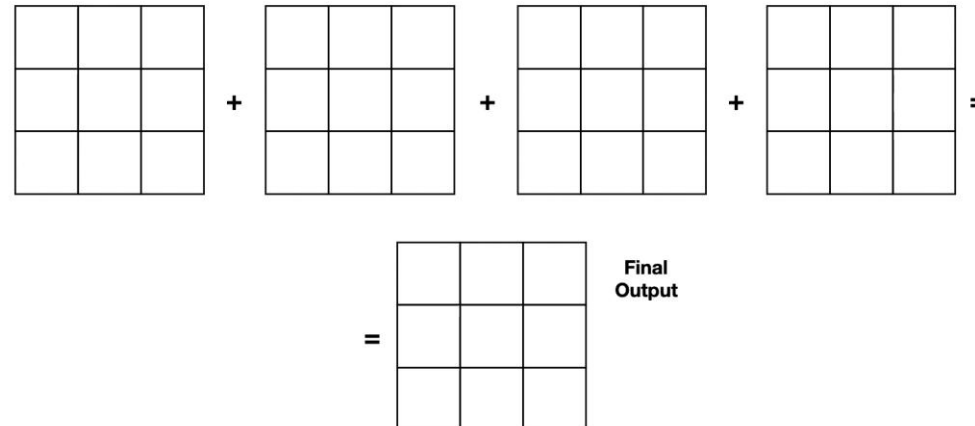
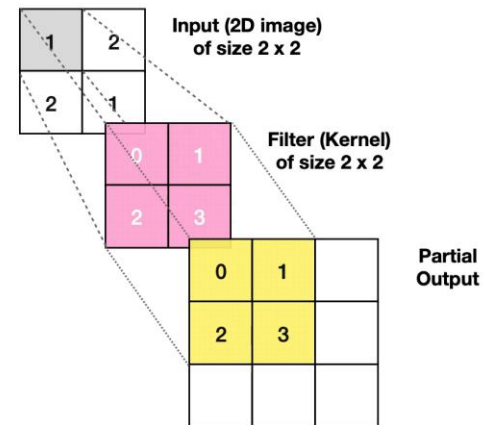


Recordemos

- Convolución transpuesta
- Pooling

Conceptos clave

Convolutacional transpuesta



<https://towardsdatascience.com/transposed-convolutional-neural-networks-how-to-increase-the-resolution-of-your-image-d1ec27700c6a/>

Datasets (e)

Pascal2

- Desafíos que impulsaron la investigación
- Inspiró el desarrollo de datasets más grandes
- Estableció un estándar en evaluación de modelos



Datasets (e)

IMAGENET



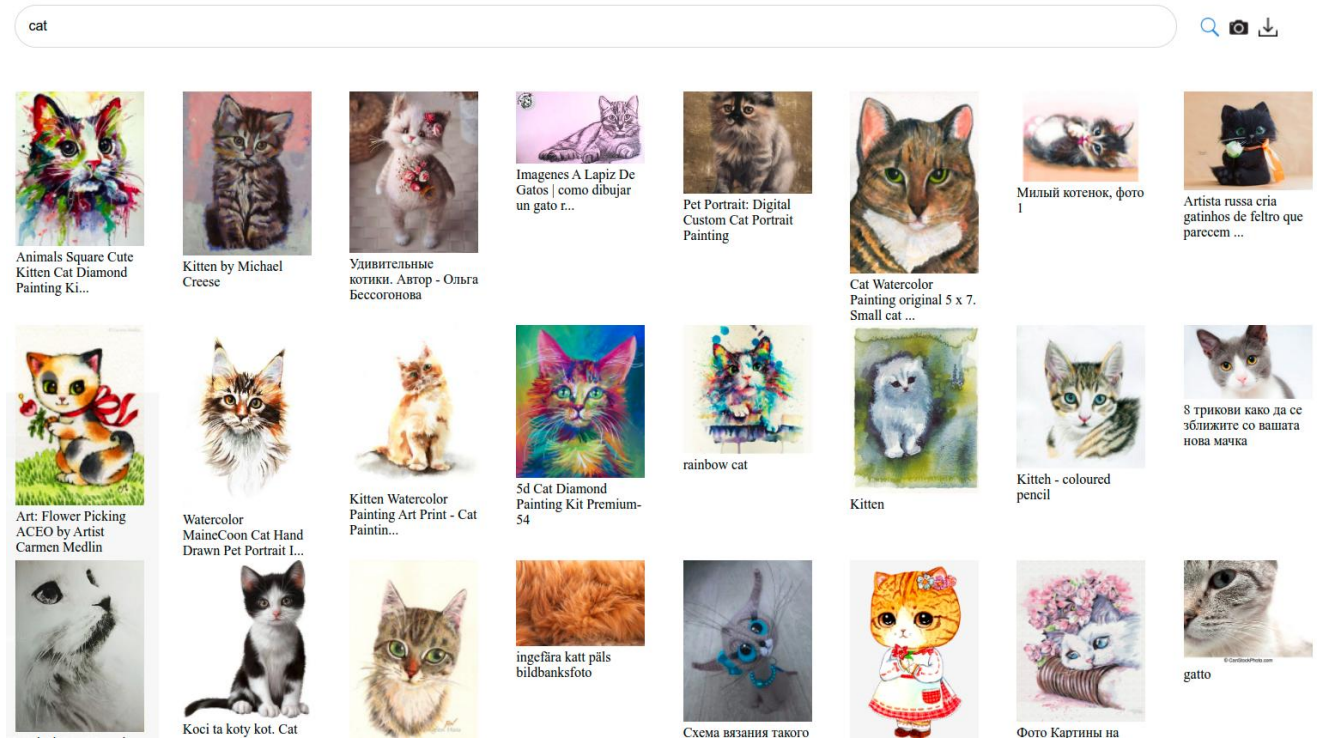
Datasets (e)

COCO



Datasets (e)

LAION 5B



Tipos de problemas

Segmentación

Clasificación

Detección

Estimación de pose

Generación de imágenes

Tipos de problemas

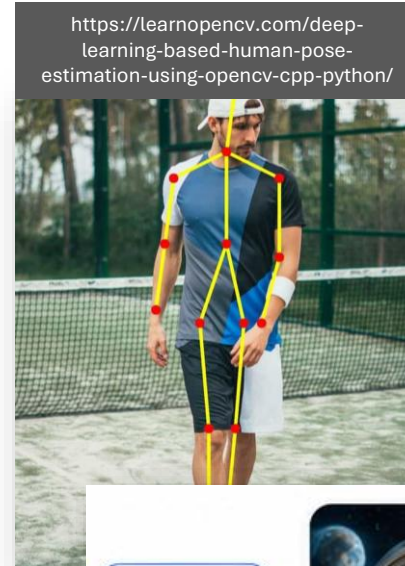
Segmentación

Clasificación

Detección

Estimación de pose

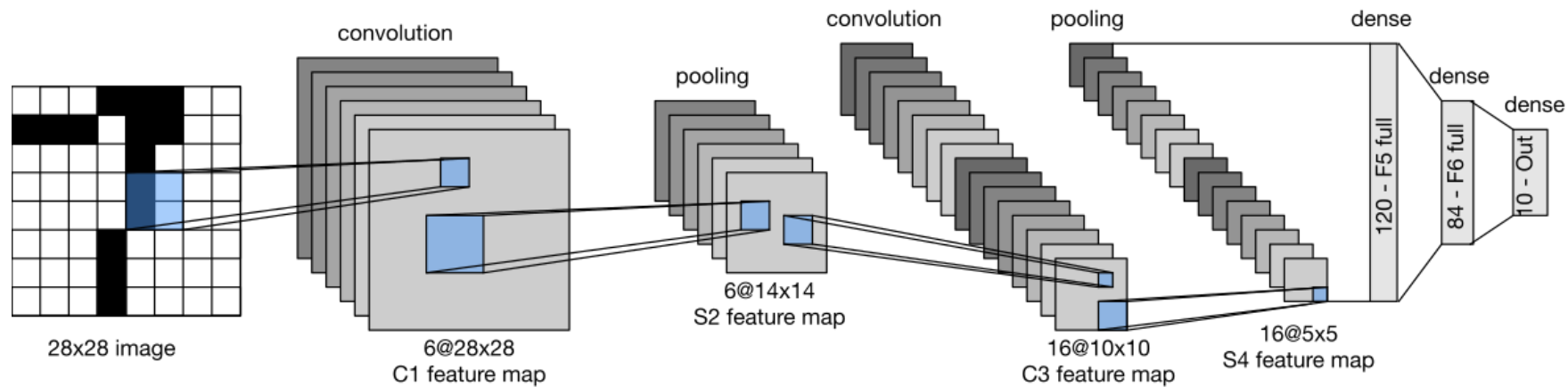
Generación de imágenes



Clasificación de Imágenes

LeNet 5

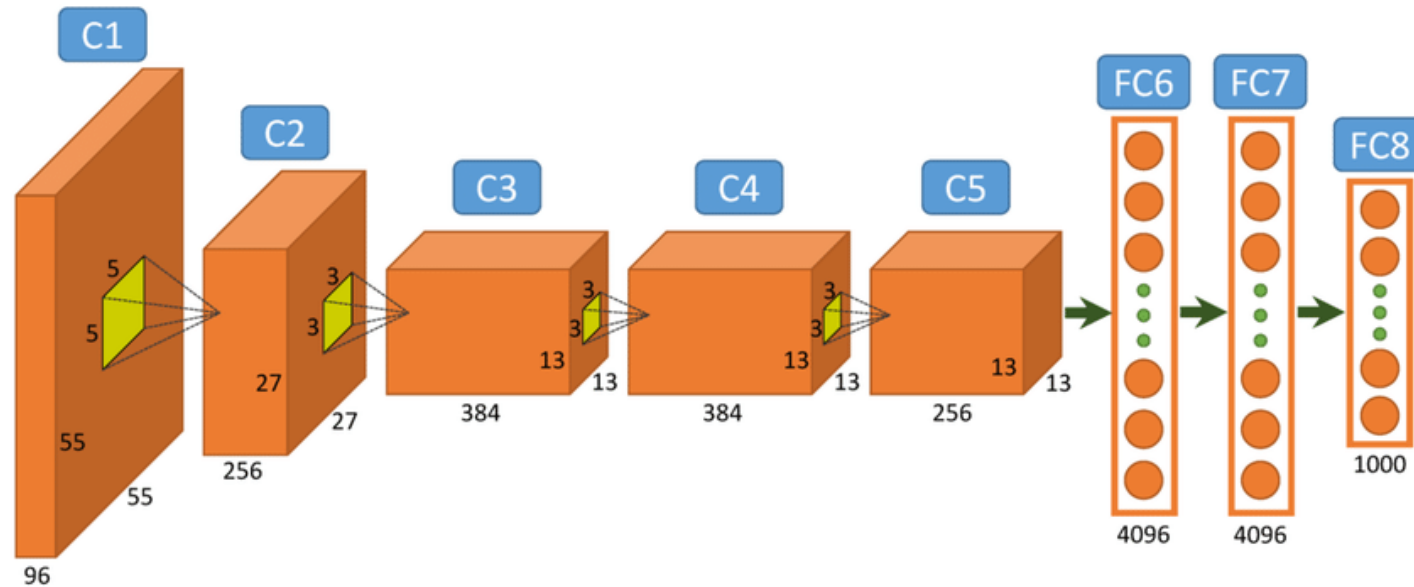
1989-1998



<https://en.wikipedia.org/wiki/LeNet>

Clasificación de Imágenes

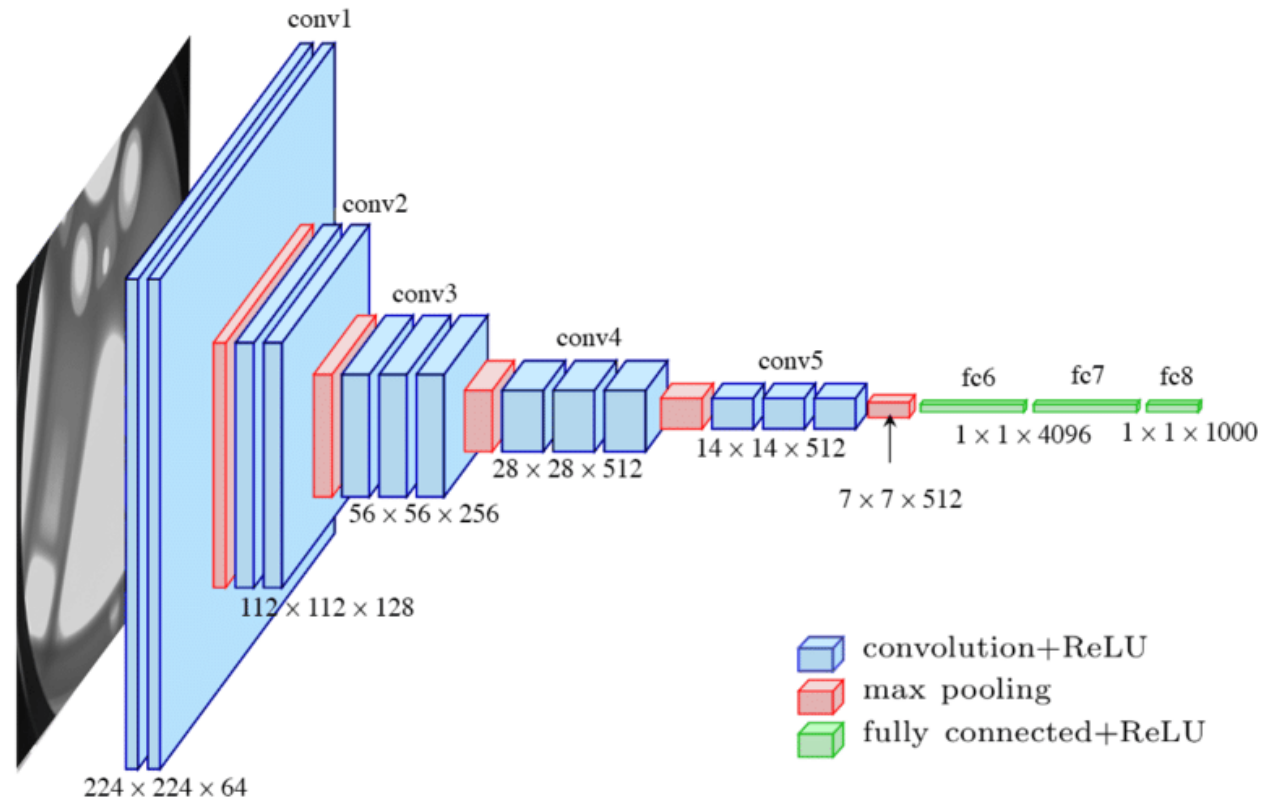
AlexNet



https://www.researchgate.net/figure/Architecture-of-Alexnet-From-left-to-right-input-to-output-five-convolutional-layers_fig2_312303454

Clasificación de Imágenes

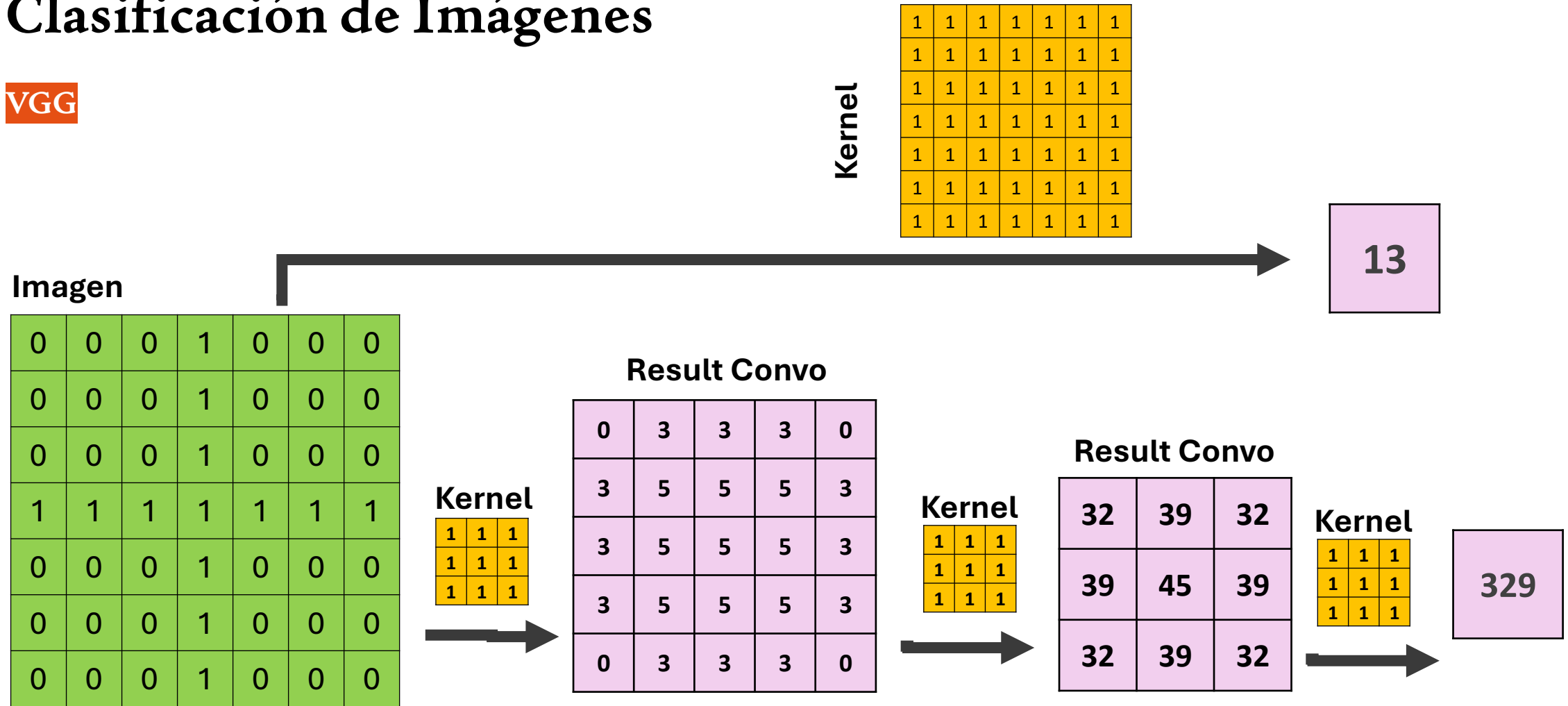
VGG



<https://developers.arcgis.com/python/latest/guide/how-ssd-works/>

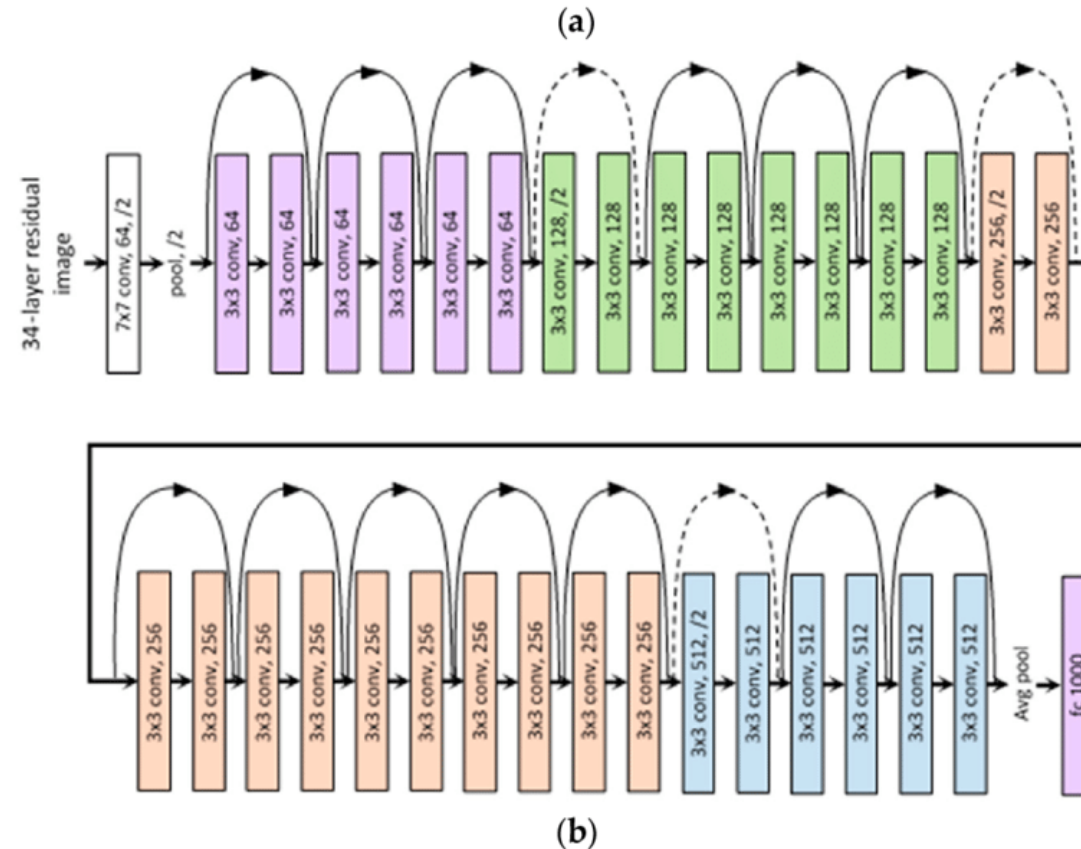
Clasificación de Imágenes

VGG



Clasificación de Imágenes

ResNET



<https://medium.com/@siddheshb008/resnet-architecture-explained-47309ea9283d>

Clasificación de Imágenes

ResNET

- Otras:

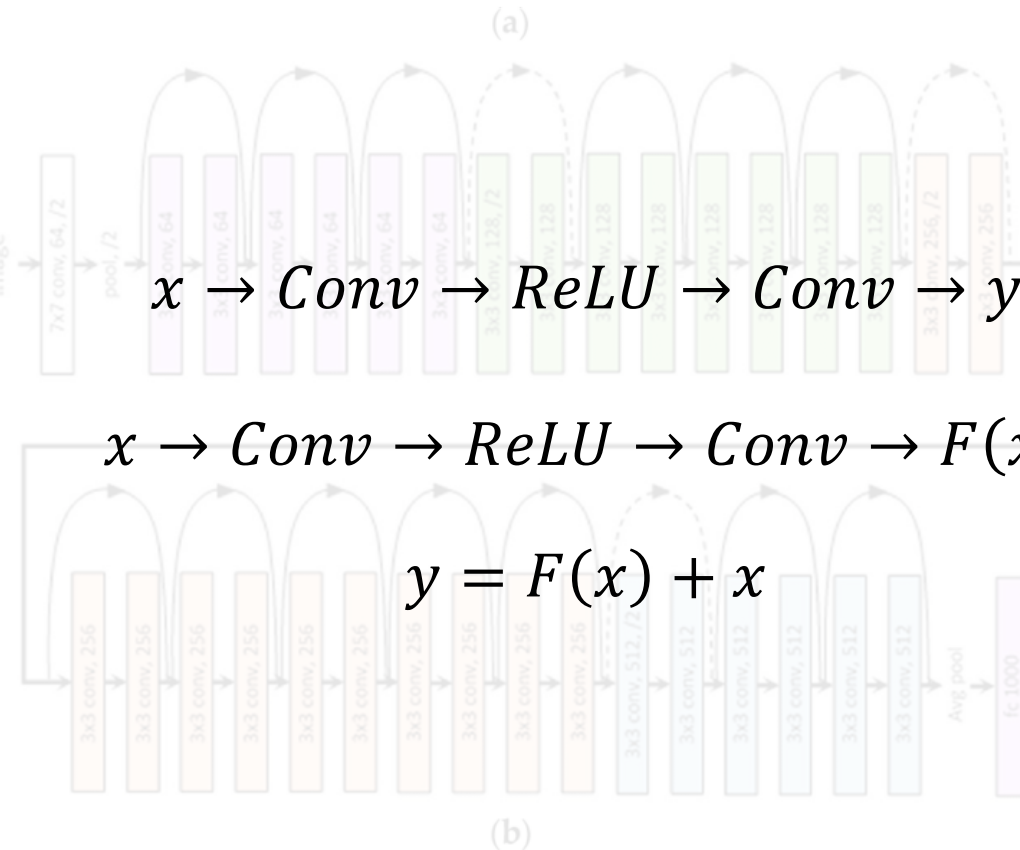
$$x \rightarrow \text{Conv} \rightarrow \text{ReLU} \rightarrow \text{Conv} \rightarrow y$$

- ResNet:

$$x \rightarrow \text{Conv} \rightarrow \text{ReLU} \rightarrow \text{Conv} \rightarrow F(x)$$

Salida

$$y = F(x) + x$$

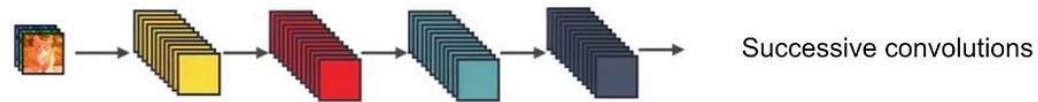


<https://medium.com/@siddheshb008/resnet-architecture-explained-47309ea9283d>

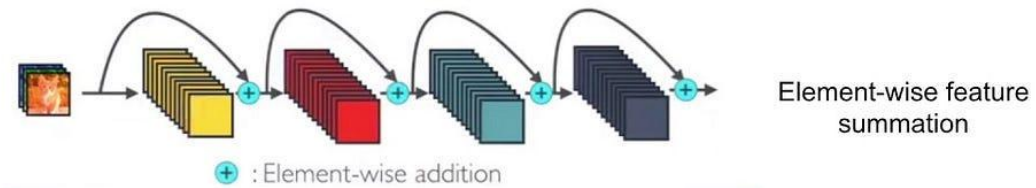
Clasificación de Imágenes

DenseNet

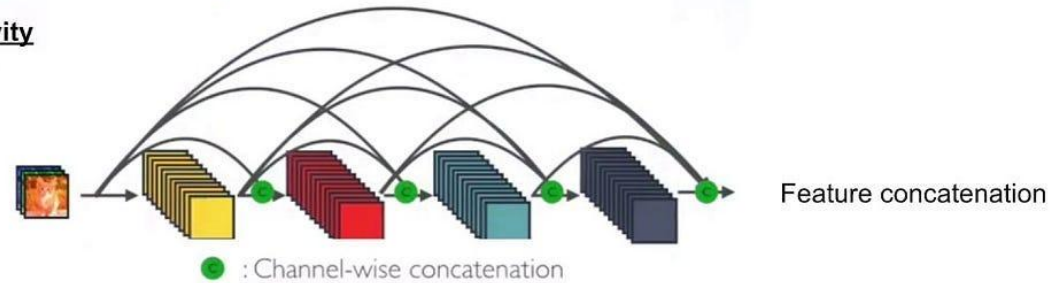
Standard Connectivity



Resnet Connectivity



DenseNet Connectivity

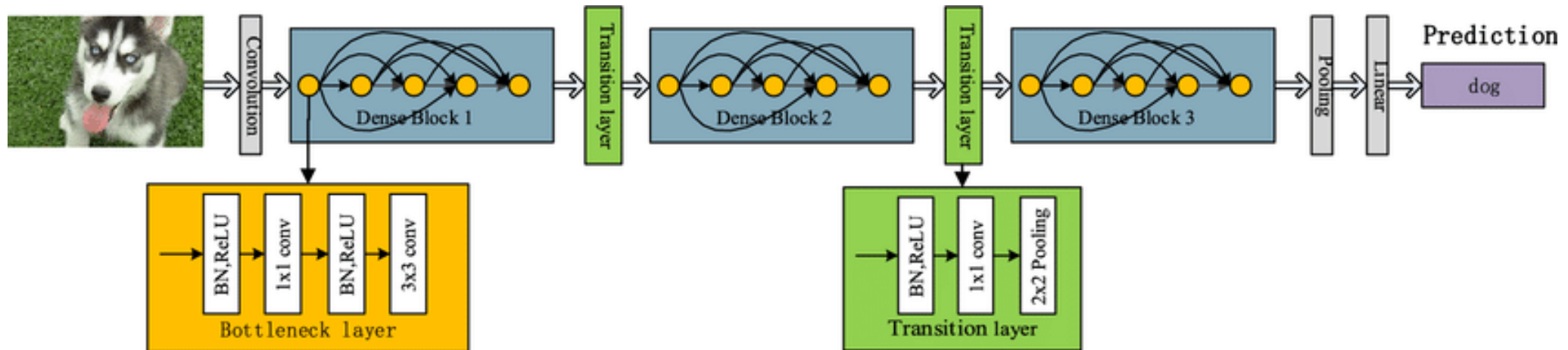


Growth rate

<https://medium.com/@abhishekjainindore24/deep-learning-architecture-7-densenet-feee44d57f89>

Clasificación de Imágenes

DenseNet

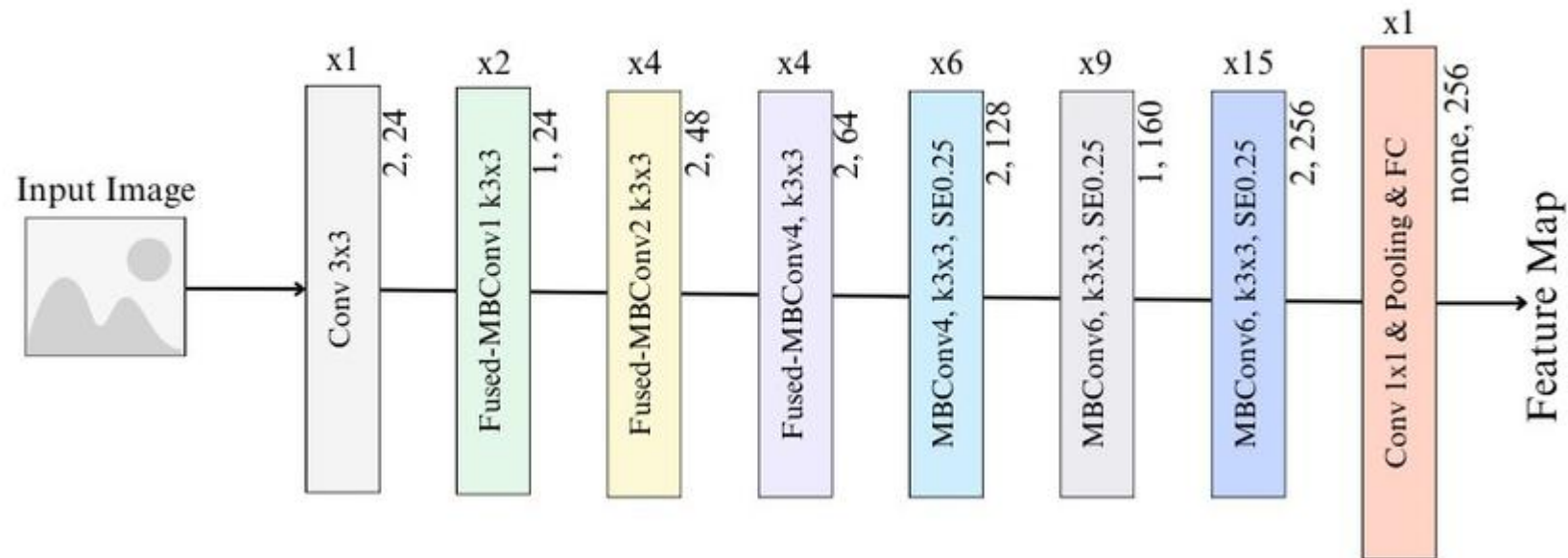


Growth rate

https://www.researchgate.net/figure/A-DenseNet-with-three-dense-blocks-and-two-transition-layers_fig1_351294032

Clasificación de Imágenes

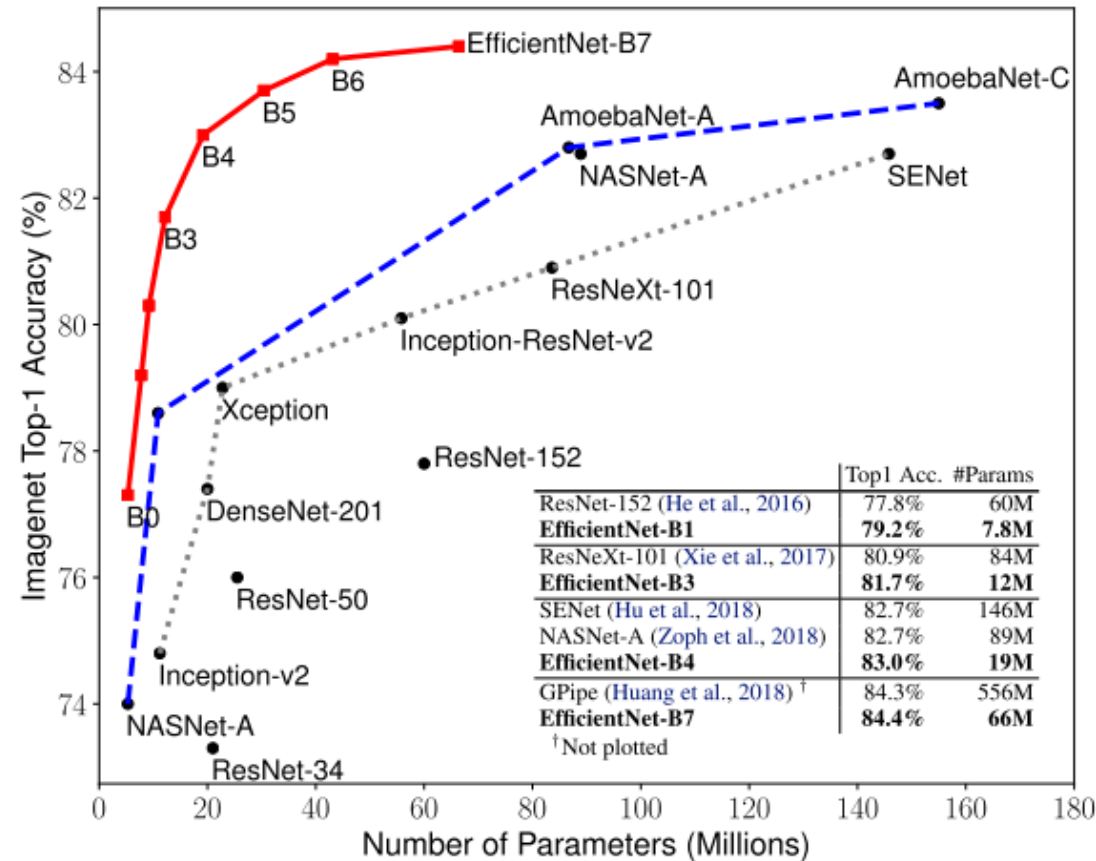
EfficientNet



https://www.researchgate.net/figure/Architecture-of-EfficientNetV2-9_fig2_379104063

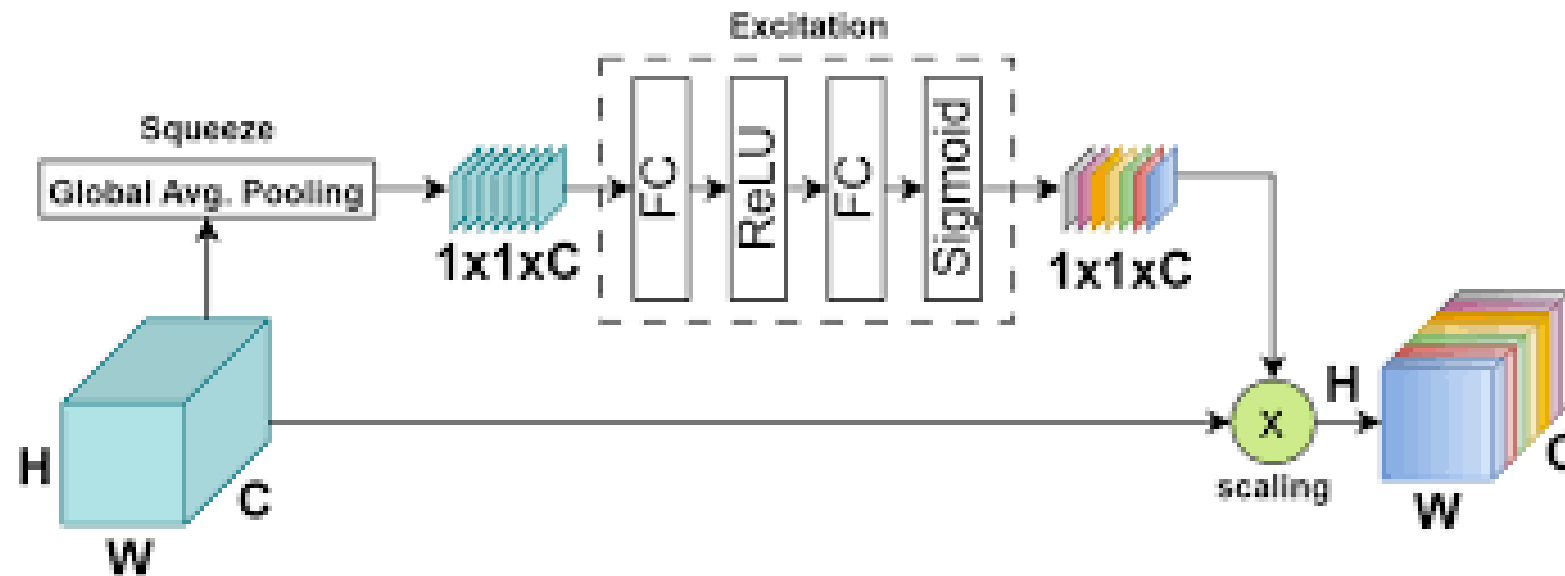
Clasificación de Imágenes

EfficientNet



<https://amaarora.github.io/posts/2020-08-13-efficientnet.html>

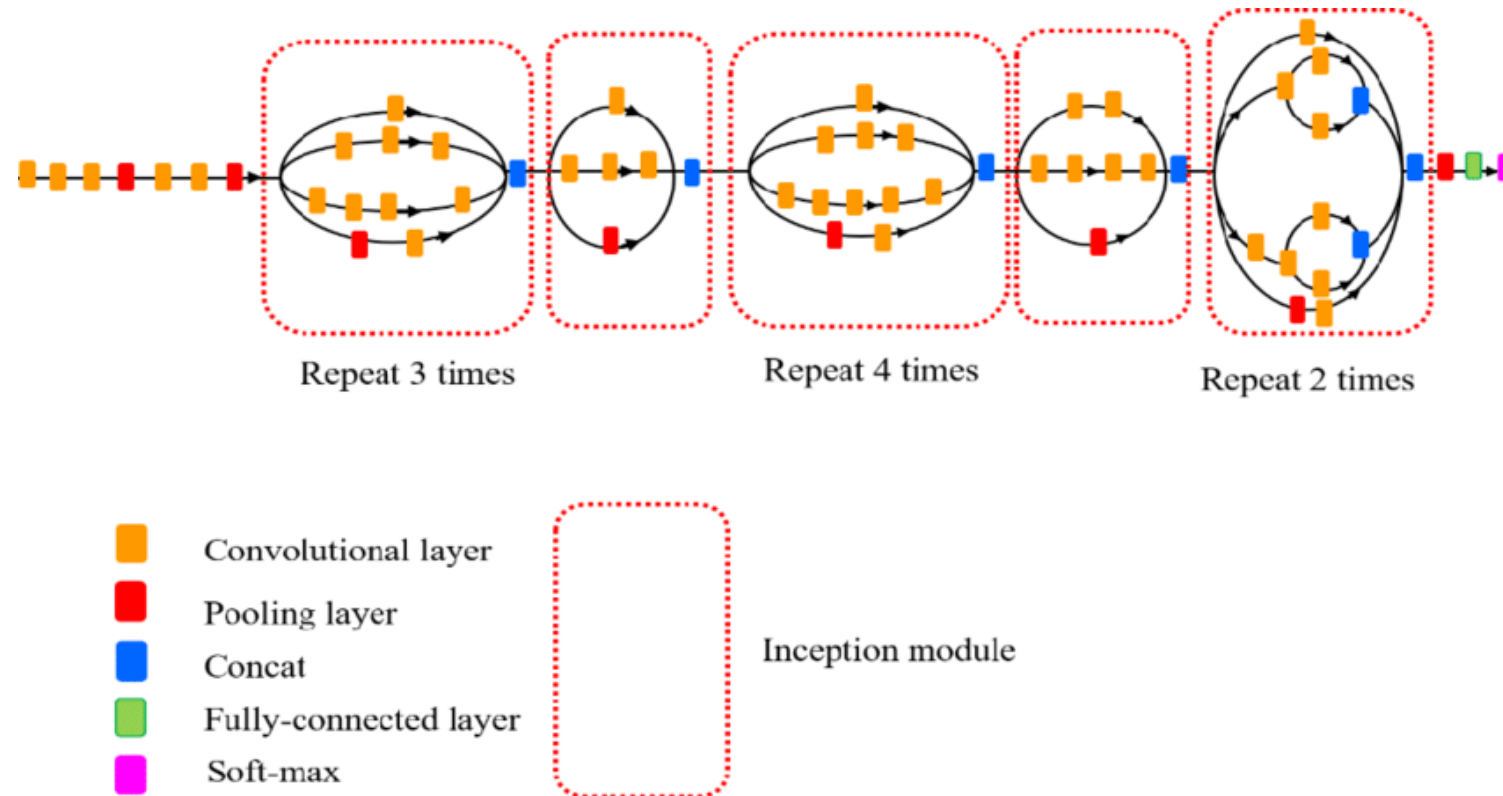
Squeeze and Freeze



<https://liora.io/es/u-net-lo-que-tienes-que-saber>

Clasificación de Imágenes

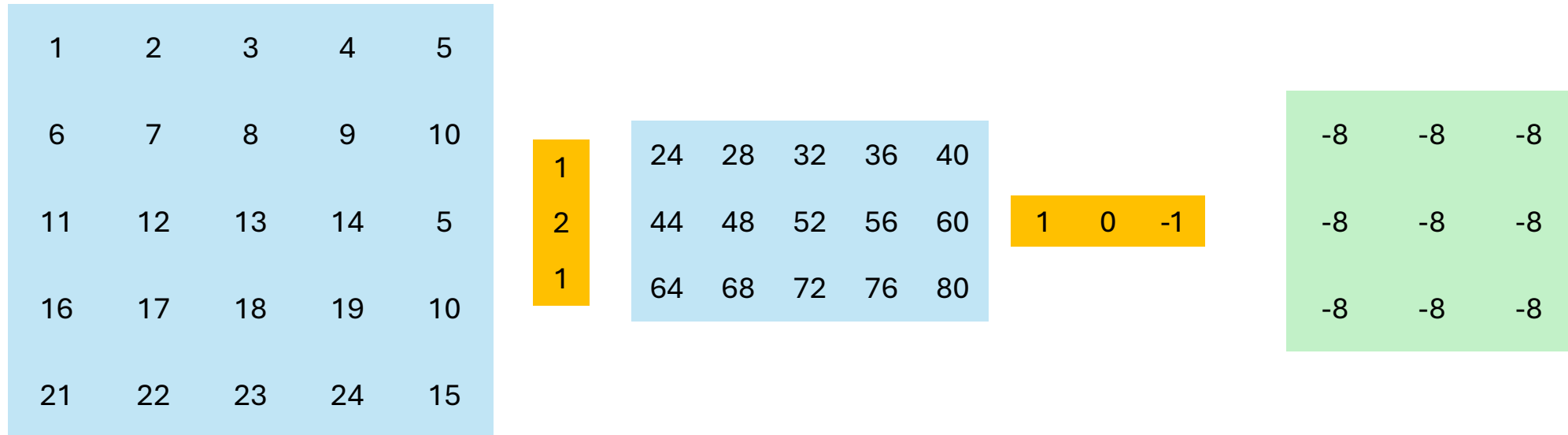
InceptionV3



https://www.researchgate.net/figure/Architecture-for-Inceptionv3-ConvNets-model-In-the-inception-modules-convolutional_fig5_327027035

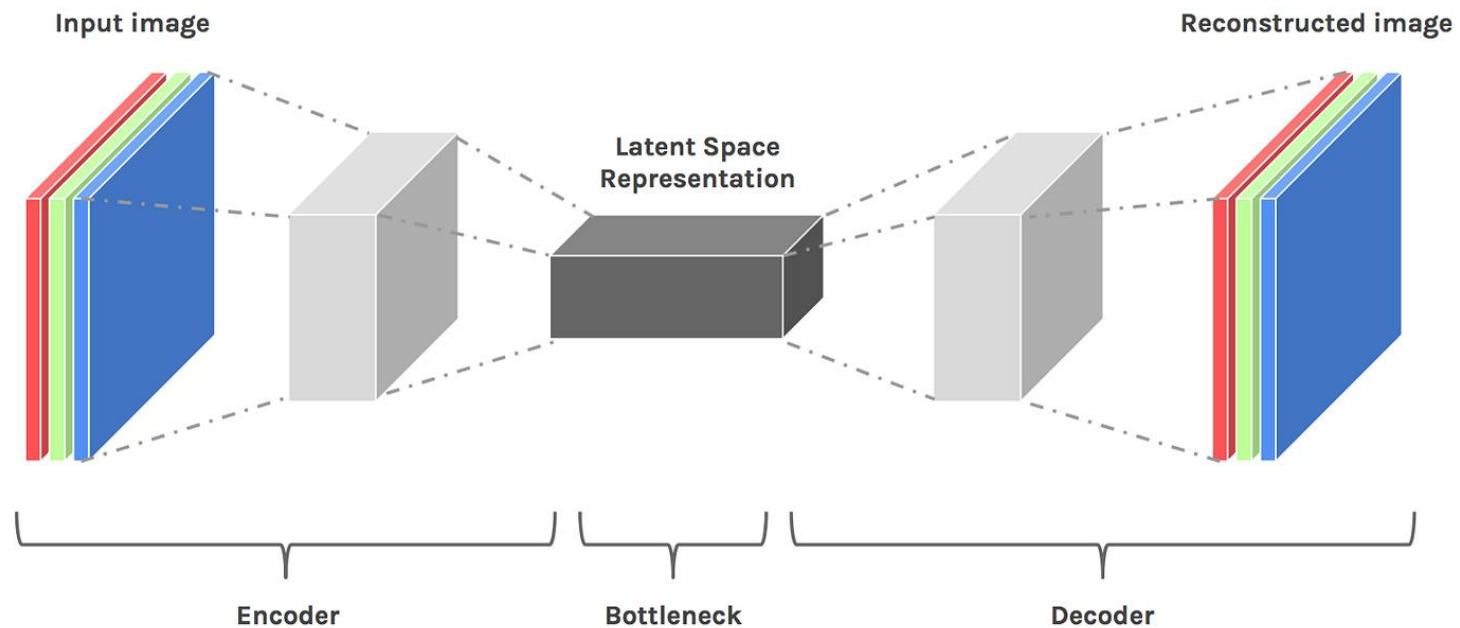
Clasificación de Imágenes

InceptionV3



Conceptos clave

Espacio Latente

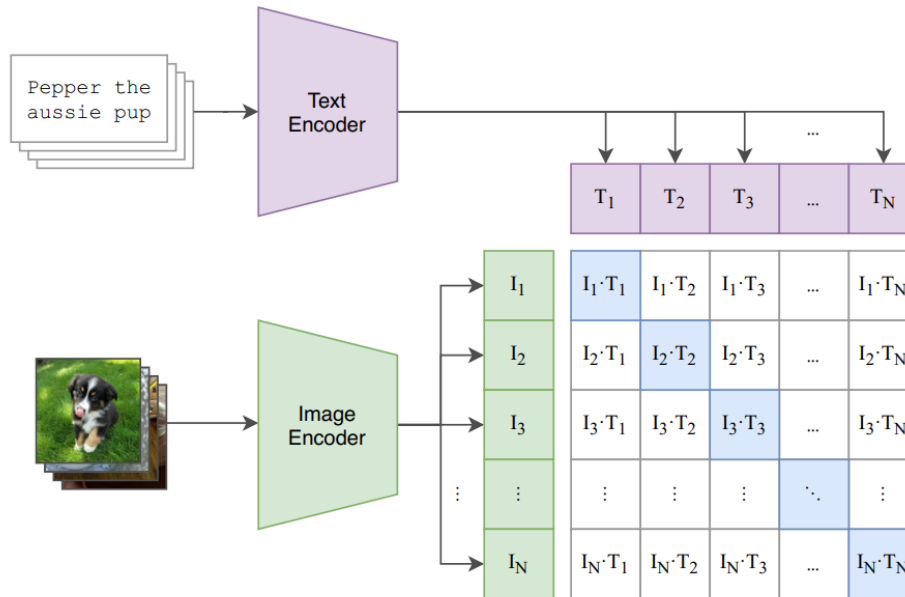


<https://medium.com/data-science/understanding-latent-space-in-machine-learning-de5a7c687d8d>

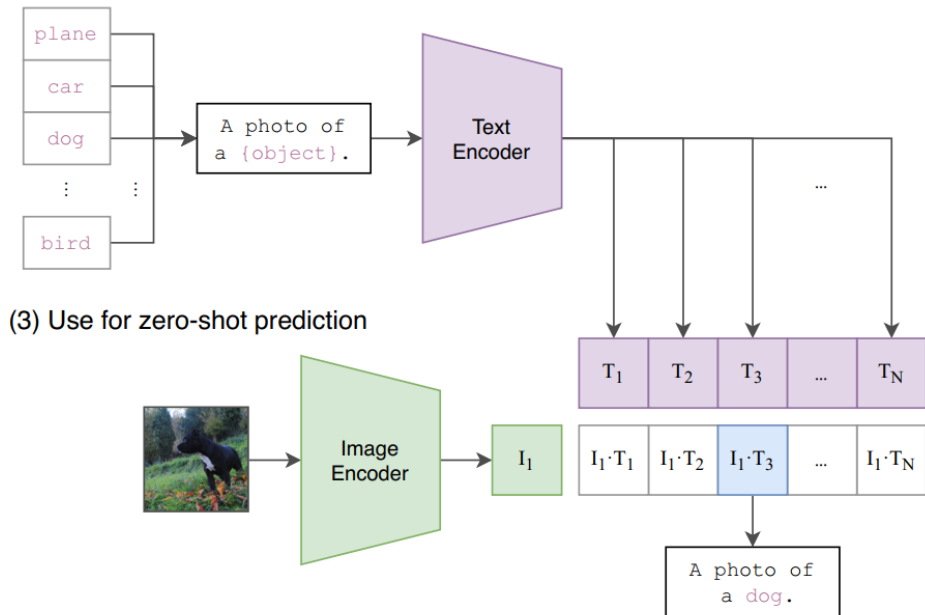
Clasificación de Imágenes

CLIP

(1) Contrastive pre-training



(2) Create dataset classifier from label text



(3) Use for zero-shot prediction

<https://towardsdatascience.com/simple-implementation-of-openai-clip-model-a-tutorial-ace6ff01d9f2/>

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